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ORIGINAL CONTRIBUTION

Suppression of Thermo-acoustic Instability inside a Rijke Tube using an Adaptive Passive Helmholtz Resonator

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Abstract Thermo-acoustic instability (TAI) inside a lean premix combustor is a major hurdle towards operation of gas turbines for both aerospace and land based operations. It arises due to the coupling between heat release fluctuation and pressure fluctuation within the combustor. It is exhibited in longitudinal, radial, tangential and bulk mode. The Rijke Tube (RT) is a canonical example to study this phenomenon on a laboratory scale in which longitudinal modes of TAI are seen. In this work a passive approach of using Helmholtz Resonator (HR) as an acoustic damper is proposed for suppression of TAI inside the RT. An attempt has been made to suppress TAI corresponding to second mode of instabilities. To abate different modes using same HR, its tuning is proposed by changing cavity volume with a hinge and use of different neck orifice diameters. The designed HR is fixed at a certain longitudinal position on RT, to suppress TAI of multiple modes. To analyse effectiveness of HR for suppressing TAI, wall pressure is measured with pressure transducer. To acquire data and analyse it, data acquisition system from National Instruments and LabVIEW software is used. The obtained result shows around 16 dB reduction in sound pressure level with HR of 5 mm neck diameter at 150° angle of cavity. The designed HR will help to reduce TAI at the design stage.

Keywords Thermo-acoustics · Rijke tube · Combustion Instabilities · Helmholtz Resonator

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Abbreviations

TAI	Thermo-acoustic instability
RT	Rijke Tube
HR	Helmholtz Resonator
QWT	Quarter wave tubes
H_f	Helmholtz resonance frequency
LPG	Liquid Petroleum Gas
APHR	Adaptive Passive Helmholtz Resonator
SPL	Sound Pressure Level

Introduction

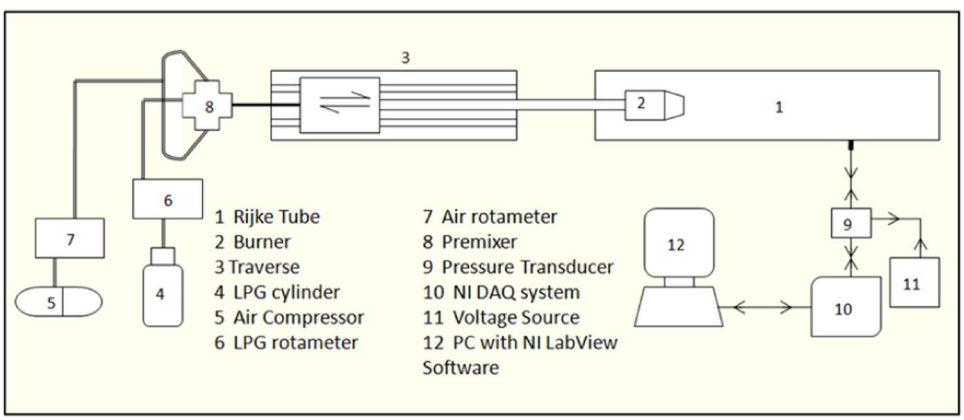
With an intention to reduce pollutants like NO_x, utilization of lean pre-mix (pre-vaporised) burners in combustors began in the decade of 1980. These pollutants were also responsible for acid rain. Unlike, the earlier combustors involving diffusion flame, these produce more power and are more efficient. Such combustors are used in gas turbines for both aerospace and land based operations. But the major problem associated with these combustors, is that they are prone to combustion instabilities or TAI (TAI) [1]. Deshmukh and Sharma [2] presented a method for TAI characterization.

TAI is self-excited resonant oscillation due to coupling between unsteady heat release and acoustic field within the combustor. An unusual humming sound or roar during the operation of a combustor implies it is subjected to TAI. This is undesirable and is accompanied with various adverse effects like excessive vibrations, high noise level, high heat transfer rates, localized burning, possibility of component melting, damage to flame holders and flame flashback [3–5]. Heat release dynamics, duct acoustics and flow dynamics are main elements of TAI. Combustion chambers can be viewed as organ pipes in which acoustic pressure and velocity oscillations can be sustained [6]. The

effect of heat source on the boundary of TAI was studied [7]. The acoustic waves generated due to monopole heat source interfere with the reflected ones forming standing waves. These resonant oscillations exhibit pressure variations near about 1% of static pressure and couple with heat release forming a positive feedback loop. In extreme cases this static pressure may go high up to 50 bar [1]. Factors like vortex shedding, equivalence ratio fluctuations, phase difference between heat release and pressure fluctuation, liquid fuel atomization with droplet evaporation, location and design of fuel injector, inlet temperature and velocity of gases are responsible for occurrence of TAI. Heat release dynamics, duct acoustics and flow dynamics or hydrodynamics are main elements of TAI [8]. Rayleigh found that it mainly depends upon the phase between heat release and pressure fluctuations [9]. To overcome the problem of arbitrary scale and unit while quantifying the Rayleigh Index, Deshmukh and Sharma [10] proposed a coefficient of Rayleigh Index based on normalizing the acquired data.

Control of TAI can be achieved by various passive and active methods. Passive methods which don't involve use of power are low-cost with no movable parts and effective at a particular frequency. Active methods are much more complicated involving use of sensors and actuators. Apart from reducing sound pressure level, these ensure that the overall combustion properties such as emission levels, flame blow out behaviour, and flashback properties are not deteriorated. An external perturbation is introduced to the system via an actuator, which manipulates the acoustic field or the heat dissipation rate. This prevents the onset of TAI or dampens it. These are effective at low frequency. Deshmukh and Sharma [11] presented simplest open loop active control technique for suppression of TAI using radial micro-jets. The net heat gain with control of TAI increases by 21.6% [12]. Unlike passive control, active control methods are more 'adaptive', and have great potential to stabilize combustors even at off-design conditions. Important passive techniques to abate TAI include use of acoustic dampers like HRs (HR) [13–15], Quarter wave tubes (QWT) [16], perforated liners [17], absorbing material [18] and baffles [19] were investigated.

A Helmholtz Resonator, an efficient acoustic damper, consists of a cavity with short neck. Mass of fluid in the body on which HR is mounted vibrates within the neck due to pressure gradient with a frequency called Helmholtz

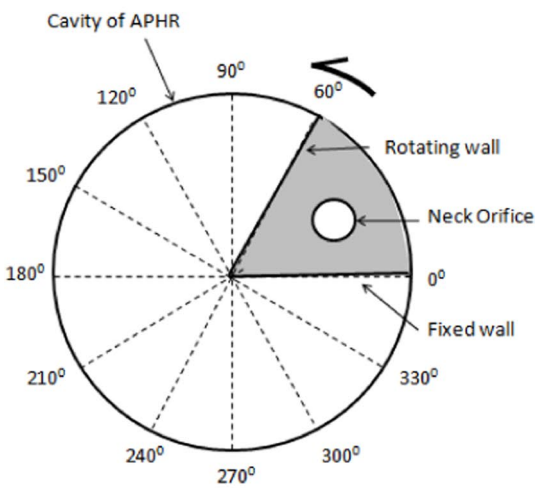


(a)

(b)

Fig.1 Experimental setup for studying TAI inside a RT **a** Schematic **b** Photograph

(a)

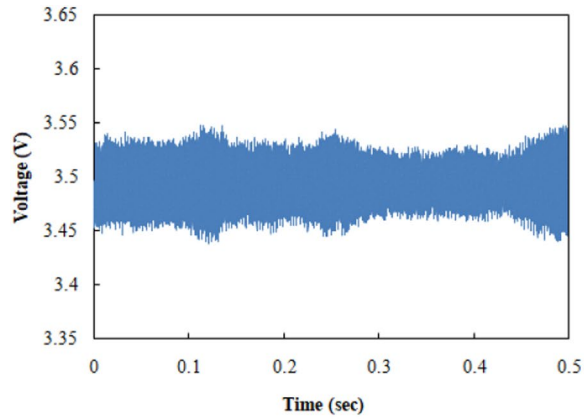


(b)

Fig. 2 **a** Photograph of APHR with hinge **b** Pictorial View

Table 1 cavity face angles corresponding to frequencies

d	θ_1 for HRF = 400 Hz	θ_2 for HRF = 900 Hz
3 mm	60.16	11.46
4 mm	107.15	21.2
5 mm	167.88	32.66



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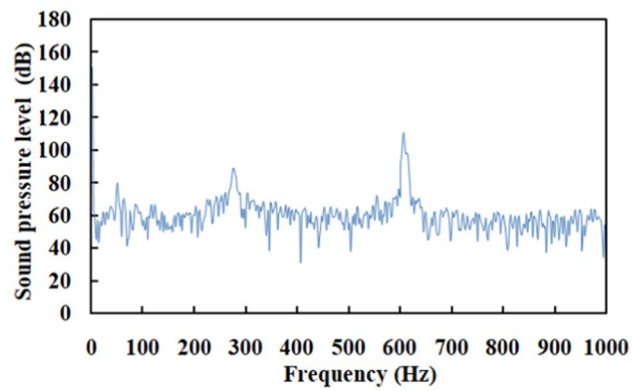


(b)

Fig. 3 **a** Captured time domain signal **b** obtained frequency spectra

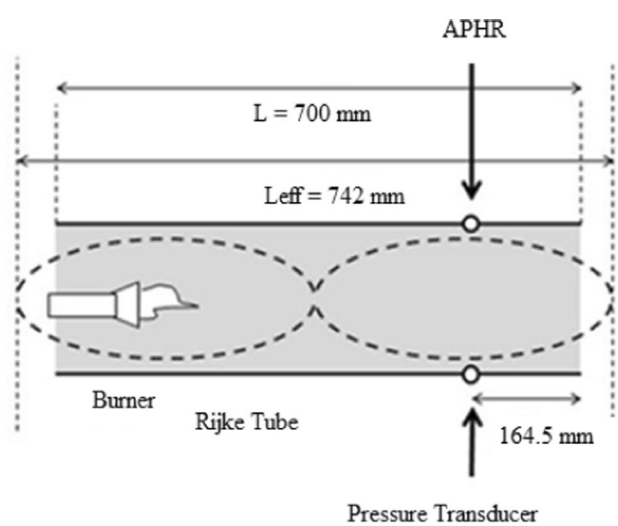


Fig.4 Placement of APHR

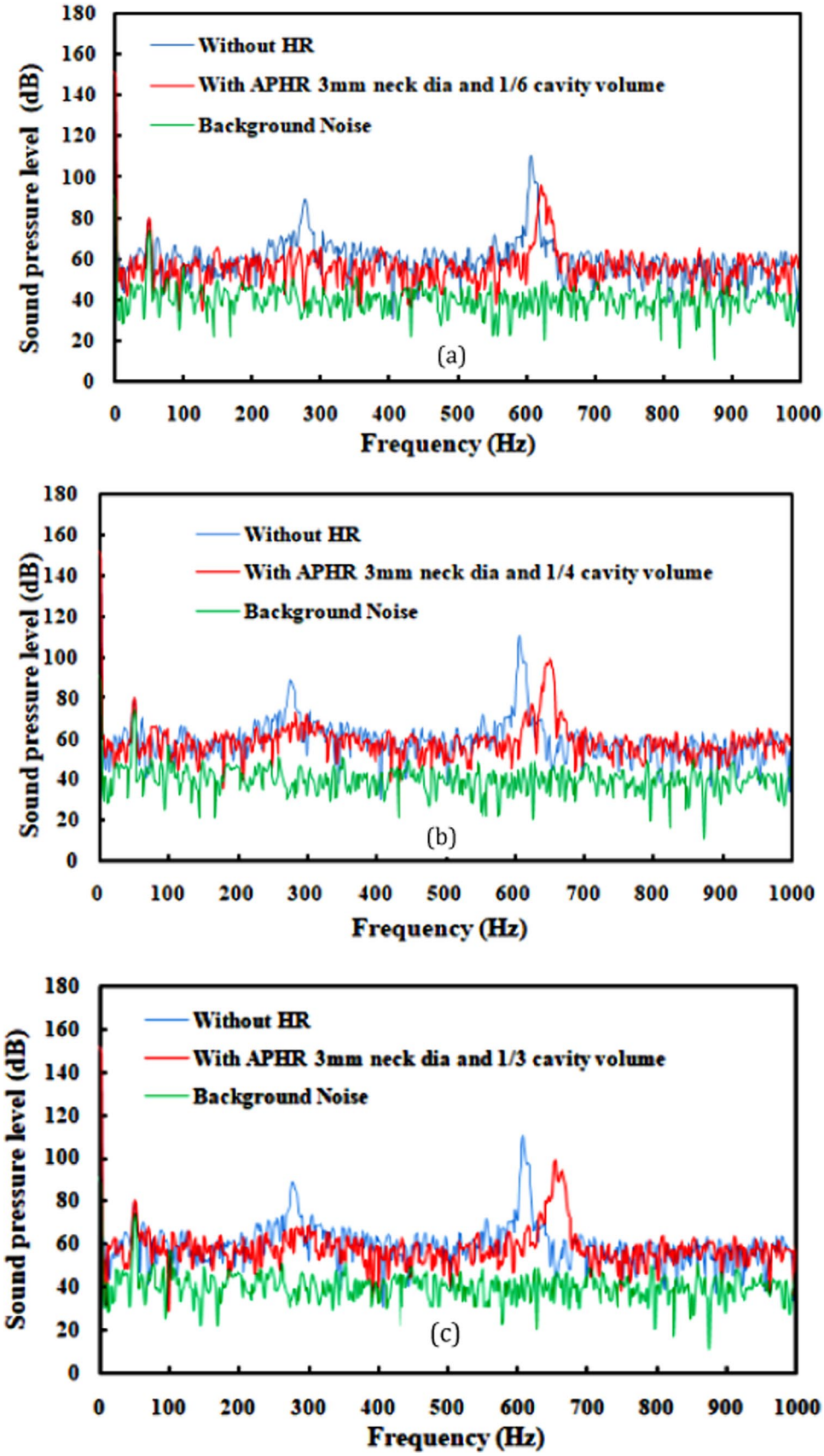
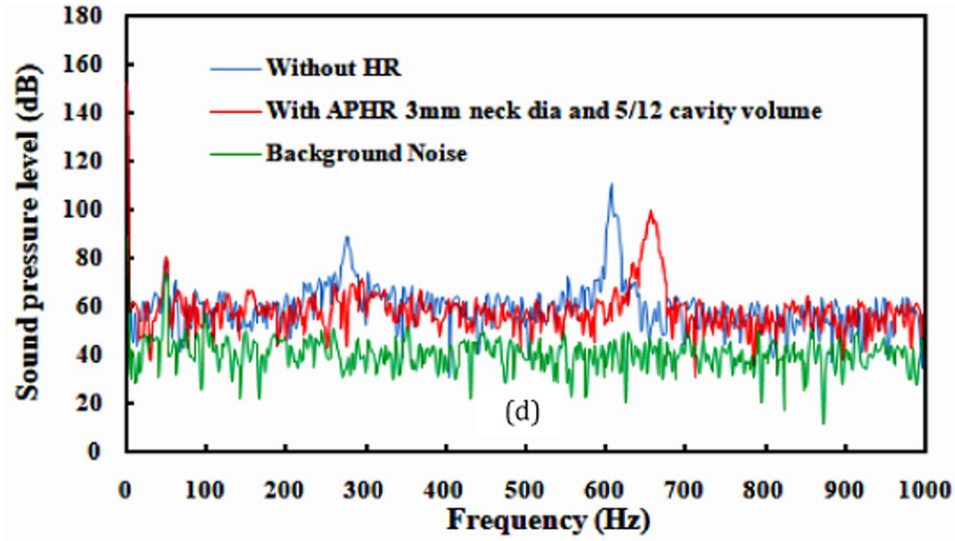
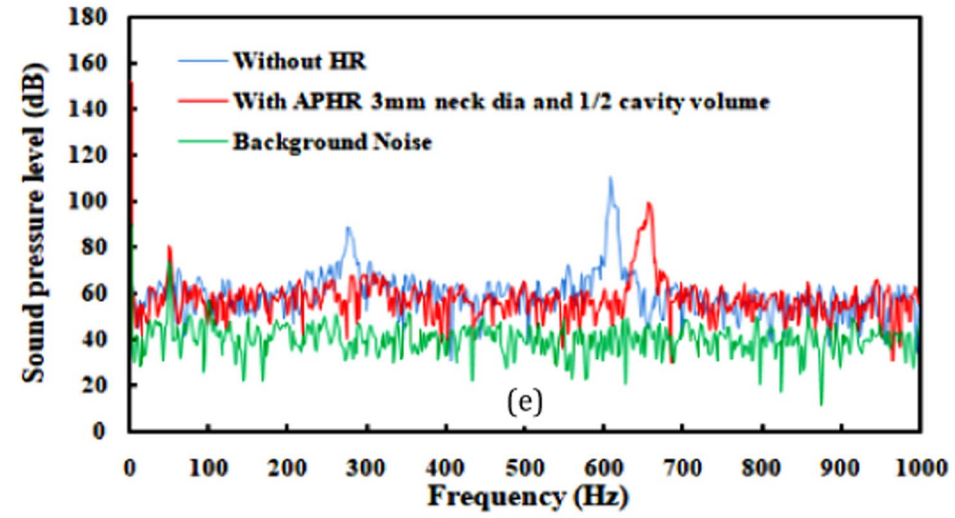
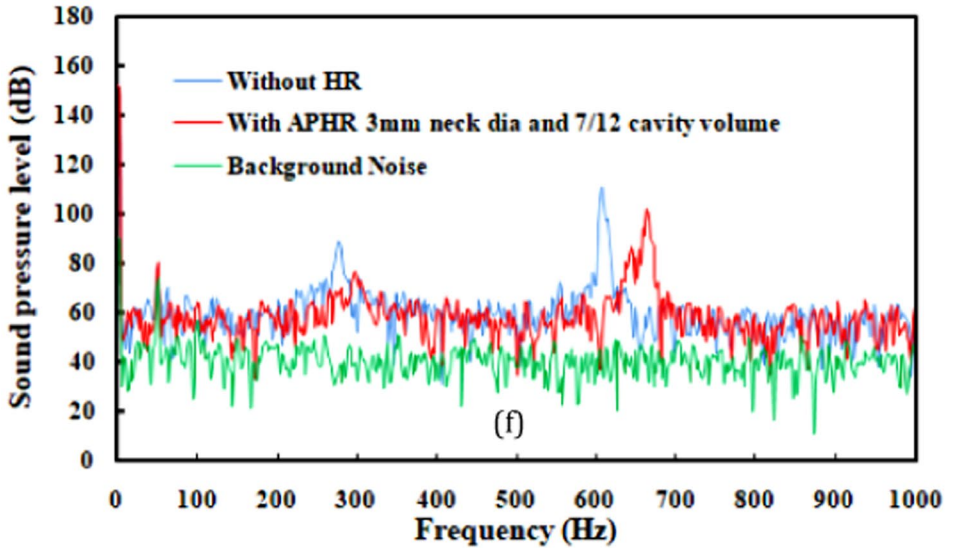


Fig.5 SPL of APHR of neck diameter 3 mm and cavity volume as **a** V/6 **b** V/4 **c** V/3 **d**5V/12 **e** V/2 **f** 7 V/12 g 2 V/3 and **h** 3 V/4

Fig.5 (continued)







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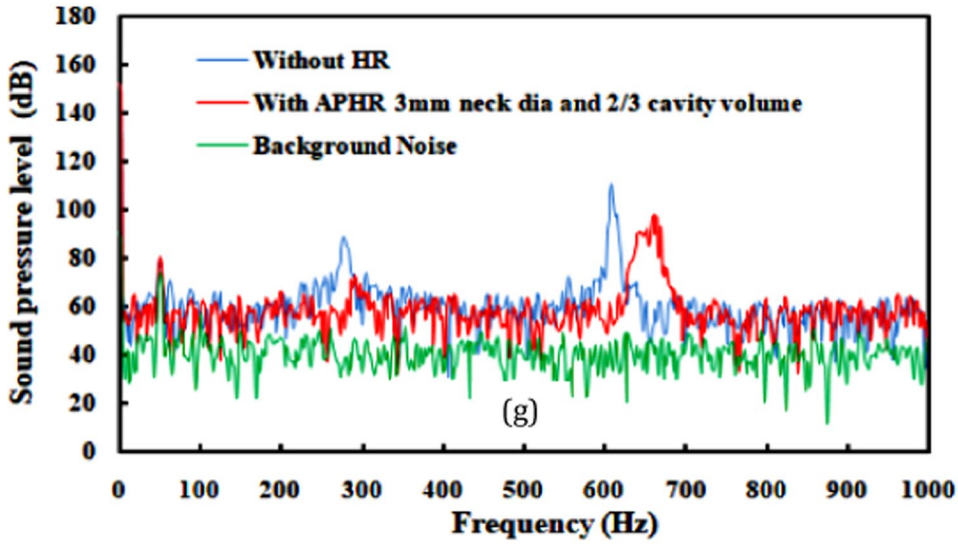
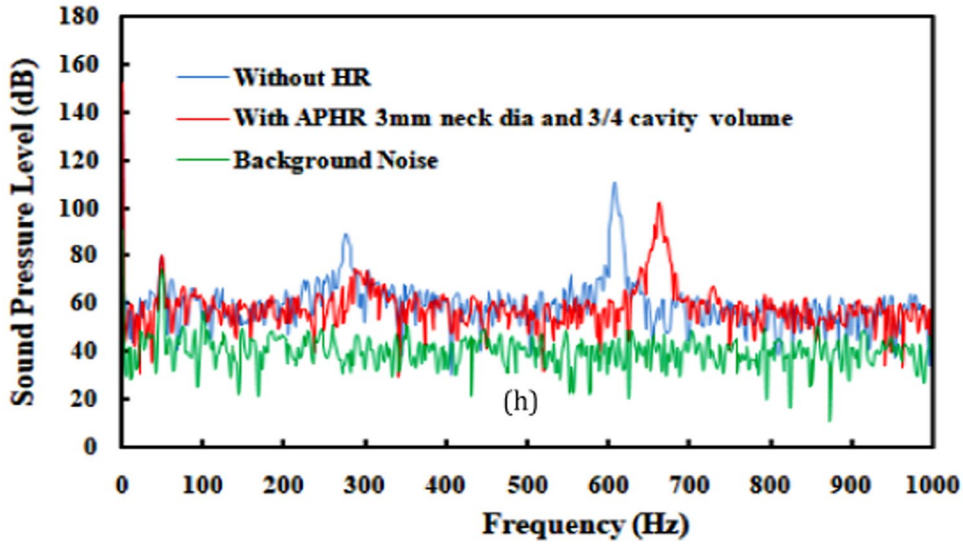




resonance frequency ($\mathbf{H_f}$) and energy is extracted. The fluid within the cavity has compliance as it gets compressed and expanded whereas boundary layer losses within the neck add damping to the system. HR is placed at pressure anti-node to extract maximum energy. When $\mathbf{H_f}$ matches with the frequency of TAI, there is reduction in Sound Pressure Level (SPL).

Bedout et al. presented algorithm to control noise using self- tuning HR [20]. Zalluhoglu and Olgac studied effects of the cavity volume, neck length and neck orifice diameter on TAI [21]. To achieve low frequency TAI attenuation and over broad frequency range, it is better to increase the cavity volume or decrease neck length or increase neck diameter [21]. Location of HR and heat source also plays a crucial role in abating TAI [22]. Krebs et al. [23] used two sets of axisymmetric HR of different sizes on annular gas turbine to dampen thermo-acoustic. Selamet and Lee [24], analyzed the effects of extended neck of HR into the cavity. It was found that such an extension acted like a quarter wave resonator in the cavity. Extension of neck into the cavity lowered the resonance frequency substantially. Furthermore, it was observed that with total extended neck length retained, neck expansion or contraction and porosity also affected transmission loss and resonance frequency [14]. The use of sound absorbing lining inside HR was analyzed and it was found that its thickness and density affect the acoustic characteristics of the resonator [14]. Keeping density fixed it was seen that increase in thickness lower transmission loss but increased stability envelop. The experimental investigations of an adaptive control for suppressing TAI in RT was carried out by Wei et al. [25].

Tunable acoustic dampers contain positive points of both active and passive approaches. It can be tuned manually or actively. It overcome major drawback of passive approaches

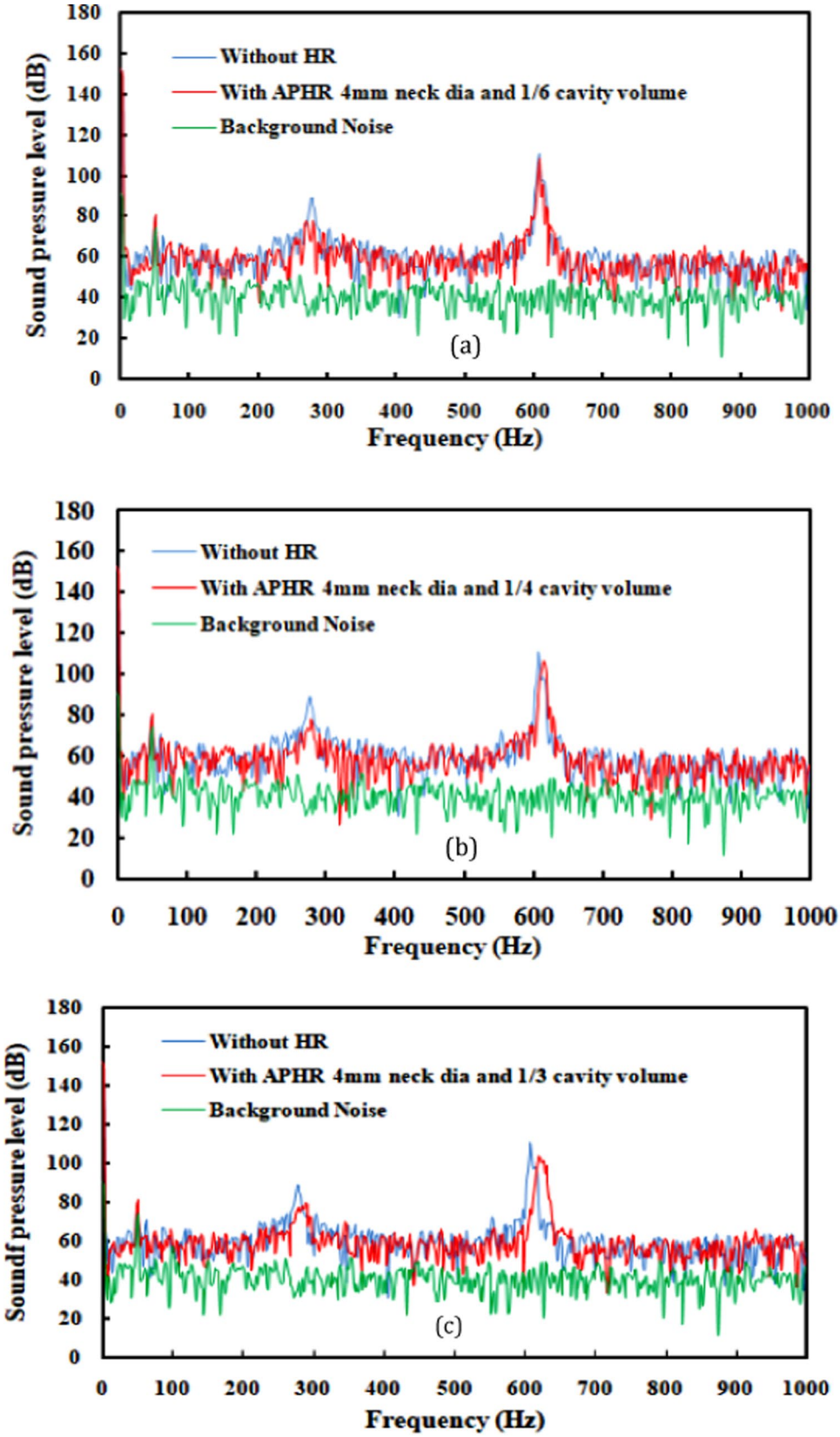
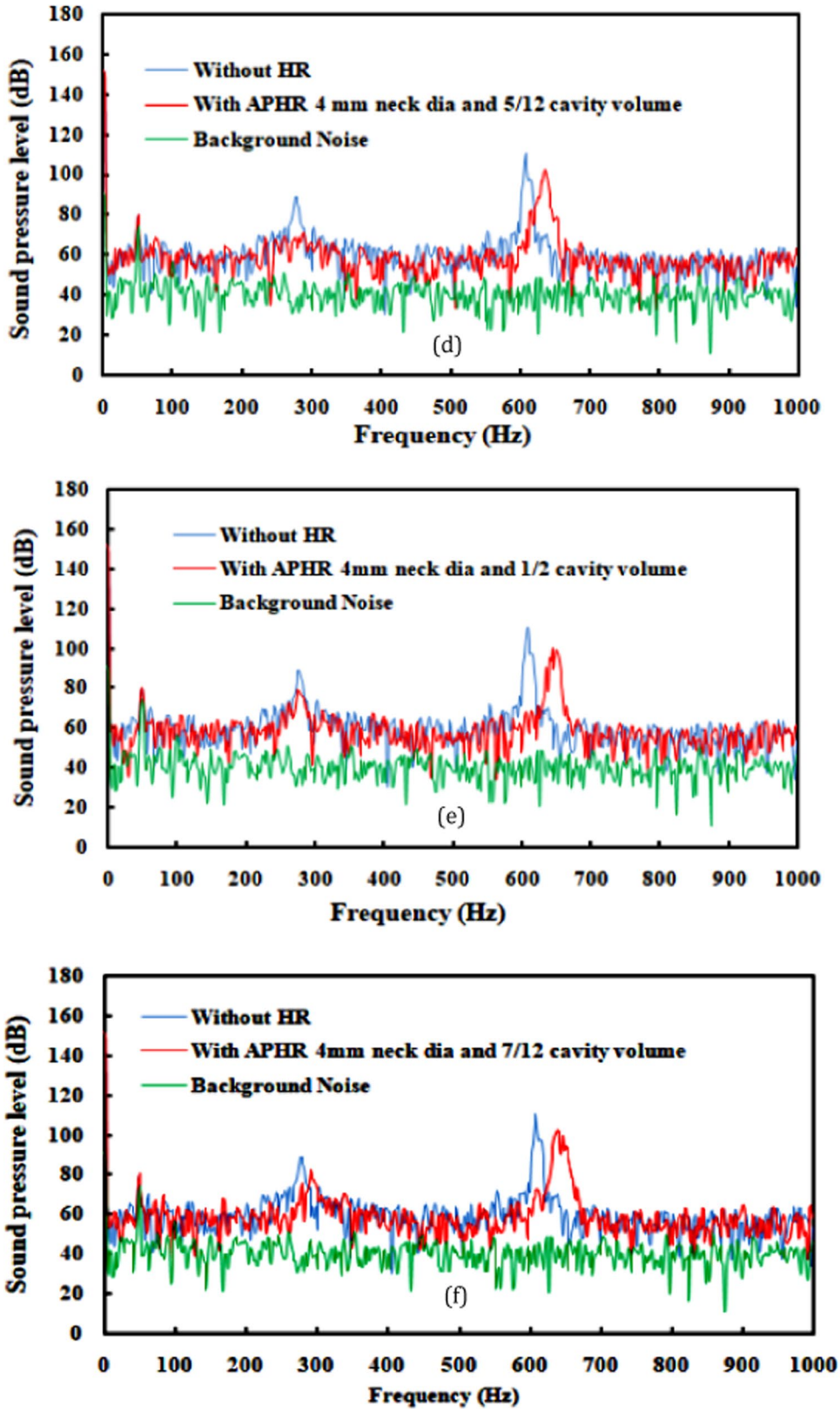


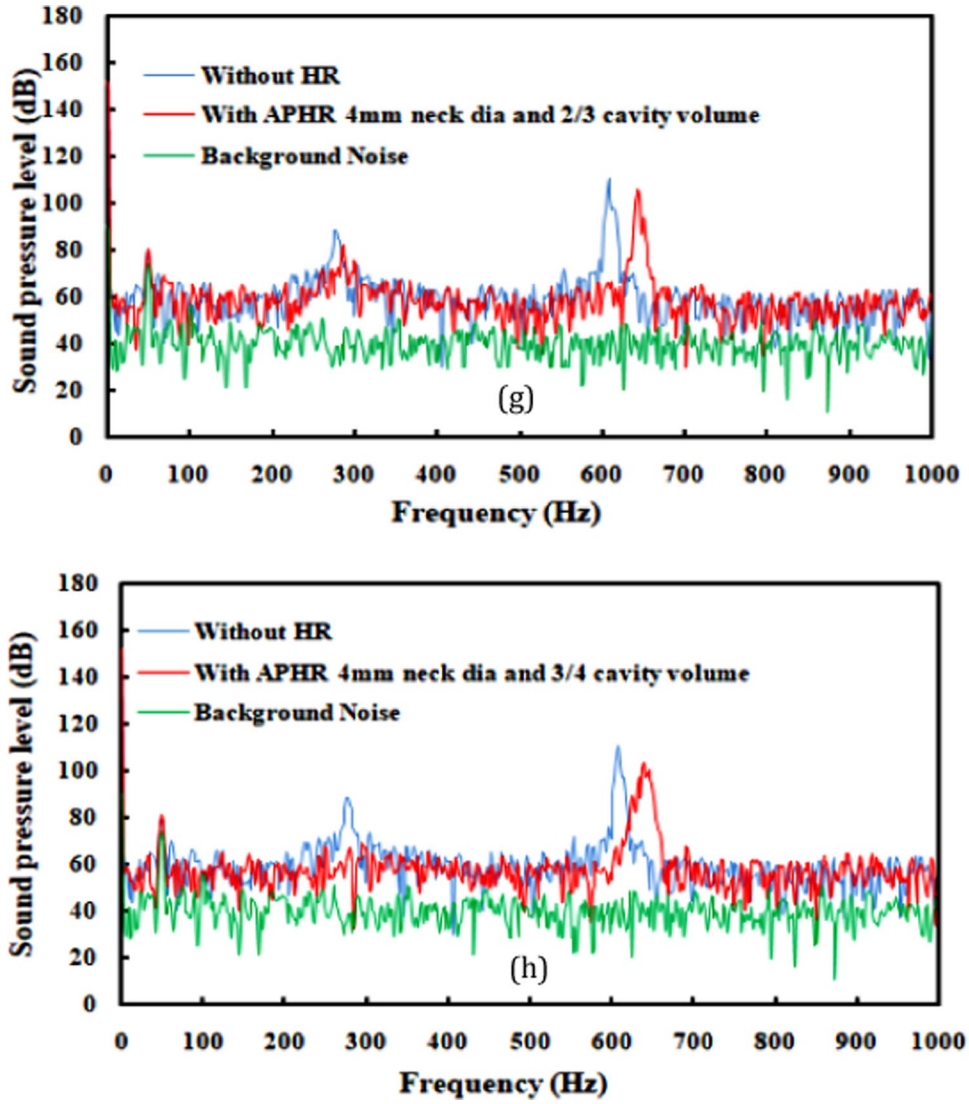
Fig.6 SPL of APHR of neck diameter 4 mm and cavity volume as **a** V/6 **b** V/4 **c** V/3 **d** 5 V/12 **e** V/2 **f** 7 V/12 g 2 V/3 and **h** 3 V/4

Fig.6 (continued)



which is effective at a particular frequency. If tuned actively, actuation only needs to be on the time scale of the changes in operating condition, which is typically much slower than the time scale of the instability. The required actuator bandwidth is therefore small, representing a major advantage over active control. Tuning of HR can be done by changing cavity

Fig.6 (continued)



volume or neck length or neck area. Seonghwi et al. [26] concluded that pure acoustic approach would be insufficient for the fine tuning of acoustic cavity, and further combustion tests would also be necessary to optimize the shapes of the cavity. If manually tuned, these are termed as adaptive passive techniques. Nagaya et al. [27] used two rotating inner cylinders to achieve changes in the HR neck area. Matsuhisa et al. [28] and Zhao and Li [29] implemented an electro-magnetic shaker and vibrating back-plate to change the volume. Helmholtz resonance frequency (**H_f**) given by Selamet and Lee [24] as shown in Eq. (1)

$$H_f = \frac{C}{2\pi}\sqrt{\frac{S}{VLeff}} \tag{1}$$

where,

C = velocity of sound.
d = diameter of neck orifice.
S = Area of neck orifice.
L_{eff} = Effective length of neck.
V = Volume of cavity.

This paper reports use of adaptive passive HR to supress, TAI of different modes. Tuning of HR is proposed by changing cavity volume with a hinge and use of different neck orifice diameters. The designed HR can be fixed at a certain longitudinal position on RT, to suppress TAI of multiple modes. The effectiveness of adaptive passive HR was evaluated for different neck diameter and volume of HRs. The proposed design of resonator can be used to suppress the TAI inside the combustor as active or passive method.

Test Setup

The experimental set up consist of a RT having length 700 mm, internal diameter 70 mm and thickness 3 mm and

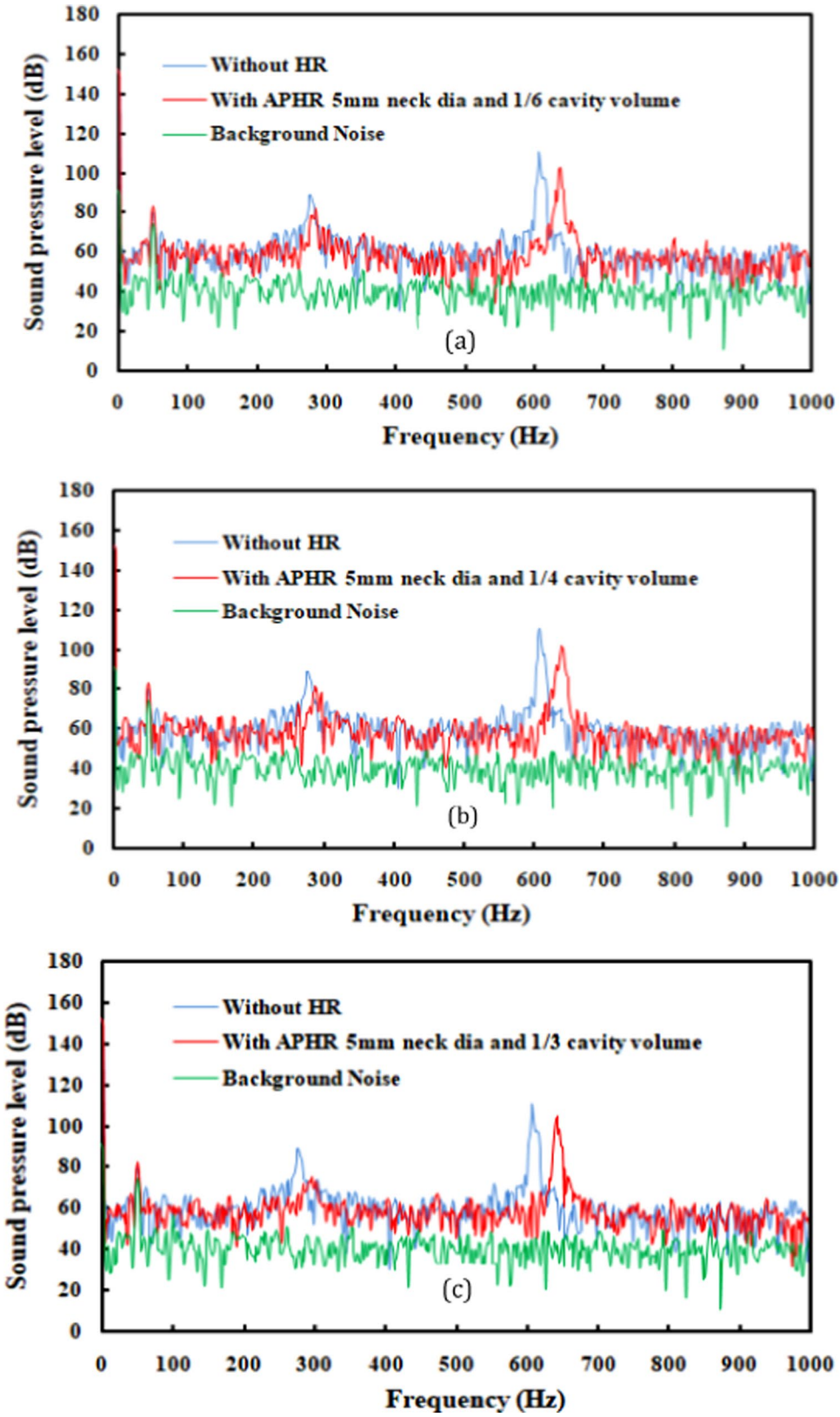
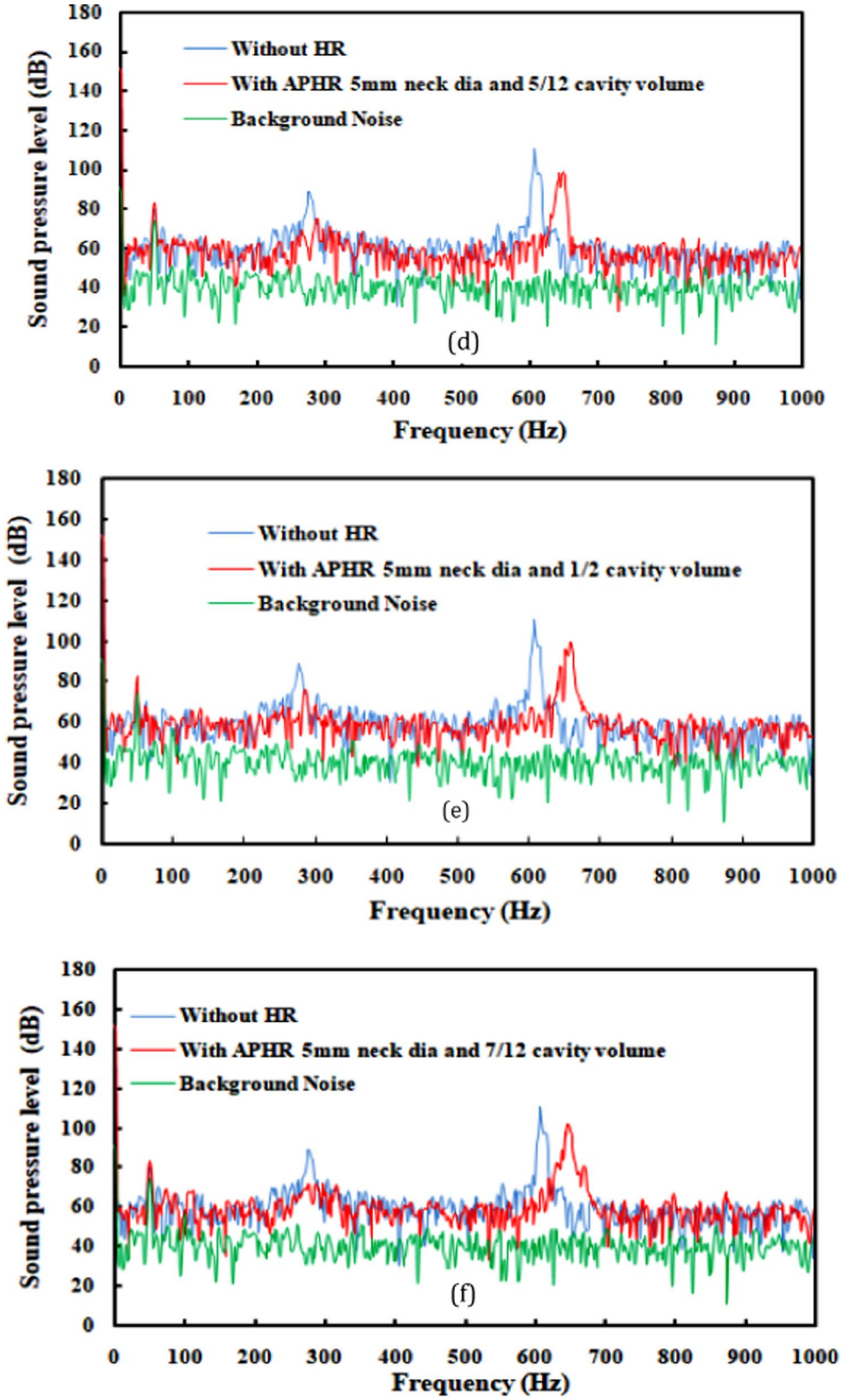


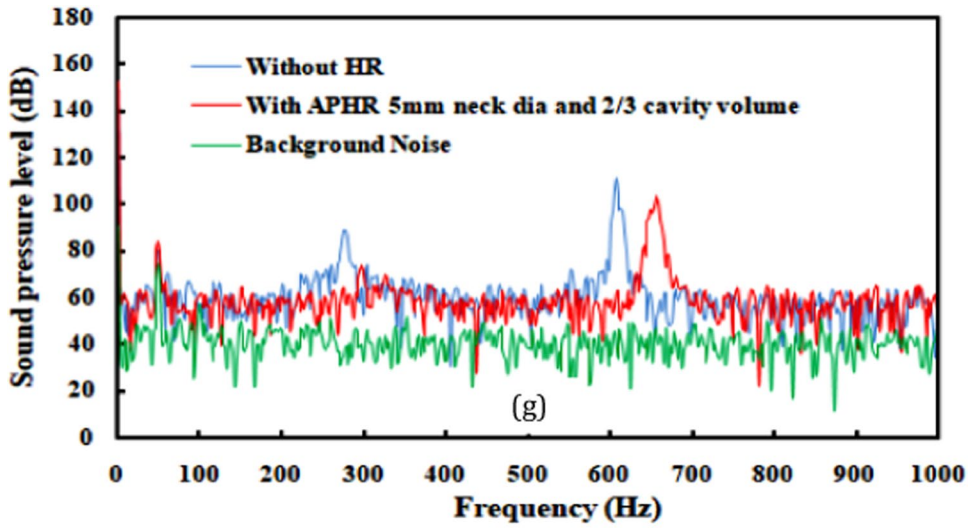
Fig.7 SPL of APHR of neck diameter 5 mm and cavity volume as **a** V/6 **b** V/4 **c** V/3 **d** 5 V/12 eV/2 **f** 7 V/12 g 2 V/3 and **h** 3 V/4

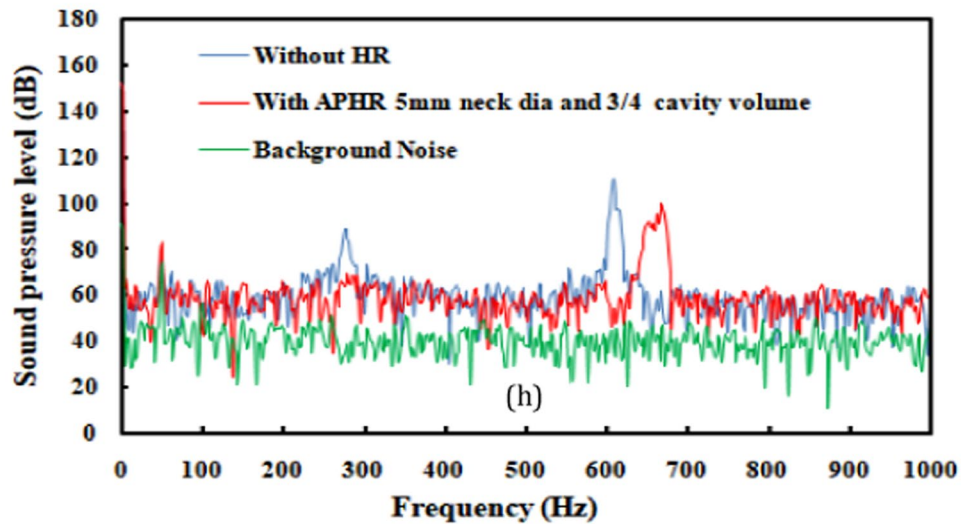
Fig.7 (continued)



is made up of mild steel. A coaxial brass burner of 30 mm diameter, connected to an inlet pipe of diameter 7 mm can move to and fro inside the tube with the help of a traverse. It has a ceramic head with sixty holes. It is connected to a pre-mixer in which air and LPG are mixed in required proportion with the help of vortex flow arrangement. The

Fig.7 (continued)





schematic of the entire setup is shown in Fig. 1a and photograph of the entire setup is shown in Fig. 1b. The flow rates of LPG and air to the pre-mixer can be changed with the help of rotameters. The range of air rotameter is 0–10 lpm with least count as 0.2 lpm whereas LPG rotameter has range 0–0.5 lpm with least count as 0.025 lpm. The fuel used in the present experiment is commercially available Liquid Petroleum Gas (LPG).

$\mathbf{L_{eff}}$ is effective length of the RT which is 742 mm considering end correction equal to 0.3D. Due to high temperature inside, corresponding thermo-acoustic modes are observed slightly higher. A burner attached with earthen head having 61 small pores is made to move inside the RT with the help of a traverse. This is connected to a pre-mixer where air and LPG are mixed to maintain air fuel ratio greater than stoichiometric ratio. A premix flame with blue colour is obtained. The instrumentation consists of data acquisition system from National Instruments model and software. The wall pressure is captured using RS-395977 Honeywell pressure transducer. An excitation of 8 V is provided to this transducer using an external power supply.

Design of Adaptive Passive Helmholtz Resonator (APHR)

This study encompasses two aspects: variation of cavity volume of HR using a hinge arrangement and use of different neck diameters. The APHR is designed in such a way that TAI frequencies in the range of 400 Hz to 900 Hz can be dampened. Second and third modes fall in this region. The cavity of this HR is formed between fixed wall attached to the cylinder and movable wall attached to the lid. The movable wall rotates about a central hinge. The volume of cavity is product of cavity face area and constant height of 46 mm. The cavity face area is in the form of a sector of radius 25.75 mm with central angle.

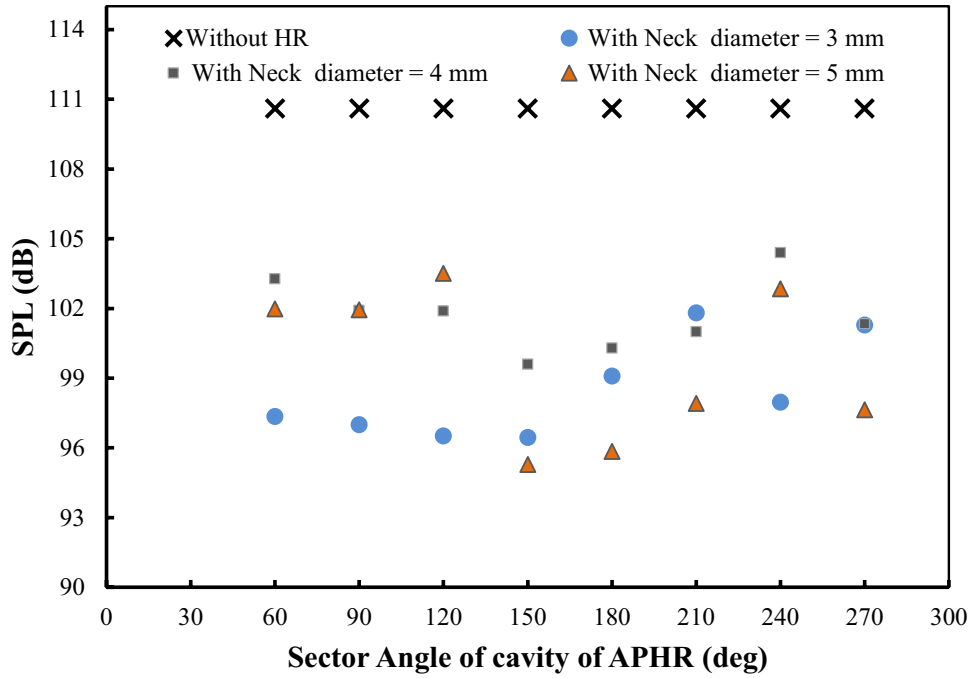


Fig.8 Variation of sound pressure level with sector angle of the APHR cavity

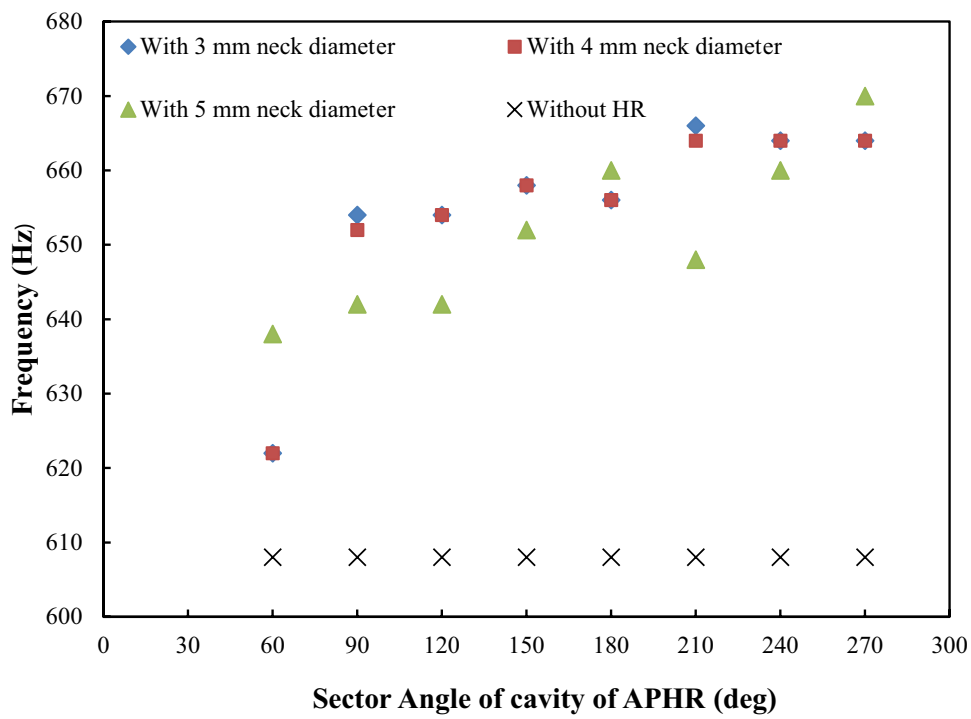


Fig.9 Variation of frequency with sector angle of the APHR cavity

This angle can be varied from 0 to 270 degrees. The photograph of Adaptive Passive HR and its pictorial view is shown in Fig. 2.

The height of cavity is less than quarter of wavelengths corresponding to frequencies lying between 400 and 900 Hz. Thus it is ensured that no standing wave formation takes place within the resonator. Different necks with orifice diameters of 3, 4 and 5 mm having same length of 11 mm are used. To ensure efficient sealing, a layer of 1 mm thick leather was applied on movable wall. Sliding parts were applied with a coating of grease. The axis of neck is offset from that of the cavity by 13 mm.

Table 1 depicts calculation of cavity face angles (θ) corresponding to frequencies 400 Hz and 900 Hz for different neck orifice diameters. Equation (1) is used to get

these space considerations. The central angle is varied from 60 to 270 degrees in increments of 30 degrees.

Experimentation

Extensive study has been carried out to identify location of burner and flow rate of burner on which strong instability occurred. Therefore, for air flow rate as 9 lpm, LPG flow rate as 0.2 lpm (A/F ratio as 45 and equivalence ratio as 1.6) and burner position at x/L = 0.06(42 cm) from left end, a thermo-acoustic mode was captured. When spectral analysis of this signal was done, a dominant peak at 608 Hz was observed. This FFT signal was then post processed to get SPL in frequency domain as shown in Fig. 3. The fundamental, second and third classical harmonics are of this RT are 272 Hz, 543 Hz and 814 Hz respectively. The frequency 608 Hz is near to 543 Hz and less than 814 Hz. Hence it is clear that acquired FFT signal is second thermo-acoustic mode. To dampen this mode, this APHR was placed at antinode point on the RT as shown in Fig. 4.

Result and Discussion

With TAI prevailing within the RT, the cavity volume was increased by rotating movable wall from 60 to 270 degrees with increments of 30 degrees. This procedure was followed for all the three neck diameters. From the acquired data frequency spectra are plotted as shown in Figs. 5, 6 and 7 for different neck diameter. To check the effectiveness of APHR, the obtained results are compared with background noise and without APHR for different volume. Considering use of APHR with all three neck diameters it was clearly seen that there is reduction in SPL with the use of APHR with a shift in frequency. The spectra clearly show, sound pressure level decrease till effective volume required for suppression and then increases. The availability of additional volume of HR not helping to supress the TAI.

The peak SPL and corresponding frequencies are captured from Figs. 5, 6 and 7. Figures 8 and 9 indicates the sound pressure level variation and peak frequency for different neck diameter and volume of APHR. Figure 8 shows the effectiveness of APHR increases (with decrease in SPL) when its cavity volume is increased from its (1/6)th part to its half. It is also clear that with use of 5 mm neck orifice diameter, maximum reduction in SPL is achieved which is about 16 dB. Volume corresponds to 150 degrees is more effective as it is closed to effective volume required for suppression. Nagaya et al. [27] proposed technique was limited to only second mode of instability, whereas the proposed method can be used for suppressing TAI of any mode. From Fig. 9, it is seen that with increase in cavity face angle, peak frequency increases due to availability of excess volume and this is observed for all three diameters. Around 60 Hz frequency shift were observed as the APHR volume increased from 60 to 270 degrees. The result obtained by Bedout et al. [20] indicates rise in frequency up to certain angle and then decrease.

Conclusion

An adaptive passive *HR* was designed and fabricated which has helped in suppression of second and third mode of TAI. This study helped in deciding the position of the resonator along the length of the RT for its effective use. It has been observed that the tuning of the resonator by variation of the sector angle results in increase of the cavity volume and hence suppression of sound pressure level (SPL). The second mode of TAI corresponding to frequency of 608 Hz has been captured and a reduction of SPL by 16 dB has been achieved, whereas study carried out by Zhao [14] able to reduce SPL by 5–11 dB. With greater neck diameter size, greater suppression of sound has been achieved as the volume of the cavity is changed from (1/6)th of total volume to half. With greater sector angle, increment in peak frequency has been observed for all neck diameters. This type of HR can be accepted as an alternative to conventional resonators which become bulky due to increase in volume with decrease in frequency. This type of HR design can be use as active control technique, if change in volume regulated through mechanism. Thus help in suppressing the instability of any mode present inside the combustor. The effectiveness of HR can be increased by using porous material.

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Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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